Chapter 3. Permutation Tests

Randomization and population models, conditional inference, permutation tests for common problems

Contents

3.1 Two sources of randomness

- **Randomization model.** N fixed units are randomly assigned, m to treatment 1 and $n = N - m$ to treatment 2. The responses are fixed numbers; only the assignment is random. Inference concerns these N units.
- **Population model.** $X_1, \dots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} G$ are random samples. Inference concerns F and G, e.g. $H_0 : F = G$.

Both lead to the same computation: the *permutation* (or *randomization*) distribution of a statistic over all re-assignments of the observed values.

3.2 The randomization test

Under H_0: "treatment has no effect," each unit's response would be the same under either treatment. The observed assignment is one of $\binom{N}{m}$ equally likely assignments, and every other assignment would have produced the same pooled set of responses. For a statistic D (e.g. $\bar{Y} - \bar{X}$), let $d_1, \dots, d_K$, $K = \binom{N}{m}$, be its values over all assignments. The upper-tailed p-value is

$$p = \frac{1}{K} \sum_{k=1}^{K} I(d_k \geq D_{\text{obs}}). \tag{3.1}$$

For a two-sided test use $|d_k| \geq |D_{\text{obs}}|$ (or a symmetric criterion), not twice the smaller one-sided value: the randomization distribution need not be symmetric.

Example 3.1

Five patients: two random patients get a new treatment with recovery times $3, 5$; the others have $8, 9, 12$. Let $D = \bar{Y}_{\text{old}} - \bar{X}_{\text{new}}$. The pooled total is 37, so for a treated pair with sum s, $D = (37 - s)/3 - s/2$, a decreasing function of s. The $\binom{5}{2} = 10$ pair sums are $8, 11, 12, 13, 14, 15, 17, 17, 20, 21$. The observed sum 8 is the smallest, so $D_{\text{obs}} = 5.67$ is the largest value and $p = 1/10$.

Proposition 3.1 (Exactness)

Under H_0, if the d_k have no ties and $\alpha = j/K$, the randomization test rejecting when $p \leq \alpha$ has type I error exactly α. With ties it is at most α.

Proof. Under H_0 the multiset $\{d_k\}$ does not depend on which assignment occurred, and the realized assignment is uniform. So D_{obs} is a uniformly chosen element of the multiset. Without ties, $p \leq j/K$ iff D_{obs} is among the j largest values, which has probability j/K. With ties, p only gets larger. ■

Proposition 3.2 (Equivalent statistics)

Over all re-assignments of a fixed pooled sample, $\sum_j Y_j$, $\bar{Y} - \bar{X}$ and the pooled two-sample t-statistic are strictly increasing functions of each other, hence give identical permutation p-values.

Proof. The pooled total $\sum X_i + \sum Y_j$ is fixed, so $\bar{Y} - \bar{X} = \frac{N}{mn} \sum Y_j - \frac{\text{total}}{m}$ is increasing in $\sum Y_j$. The total sum of squares SST is also fixed, and $SST = (N-2)s_p^2 + \frac{mn}{N} D^2$ with $D = \bar{Y} - \bar{X}$. Hence

$$T = \frac{D}{\sqrt{\frac{1}{m} + \frac{1}{n}} \sqrt{(SST - \frac{mn}{N} D^2)/(N-2)}} = g(D), \qquad g'(D) \propto \frac{SST}{(SST - \frac{mn}{N} D^2)^{3/2}} > 0.$$

■

Statistics such as a difference of medians or of trimmed means are *not* monotone in D and can give different (often more powerful, for skewed or heavy-tailed data) tests.

3.3 Permutation tests in the population model

Motivation. Randomization justifies permutation tests in designed experiments. The same tests remain valid for random samples from two populations, because under H_0 the pooled sample is exchangeable.

Definition (Exchangeability)

$(Z_1, \ldots, Z_N)$ is **exchangeable** if $(Z_{\pi(1)}, \ldots, Z_{\pi(N)}) \stackrel{d}{=} (Z_1, \ldots, Z_N)$ for every permutation π.

I.i.d. variables are exchangeable. So are equicorrelated normal variables $N_N(\mu\mathbf{1}, \sigma^2[(1-\rho)I + \rho\mathbf{1}\mathbf{1}'])$, which are not independent. Two independent samples from $N(\mu, 1)$ and $N(\mu, 4)$ are *not* exchangeable, even though their means agree.

Theorem 3.3 (Validity of permutation tests)

Let $Z = (Z_1, \ldots, Z_N)$ be exchangeable under H_0, let $\mathcal{G}$ be a group of permutations, and let $p(Z) = \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}} I(T(gZ) \geq T(Z))$. Then $P_{H_0}(p(Z) \leq \alpha) \leq \alpha$ for every α, with equality when $\alpha|\mathcal{G}|$ is an integer and there are no ties. The statement holds conditionally on the multiset of observed values, hence also unconditionally.

Proof. Let U be uniform on $\mathcal{G}$, independent of Z. By exchangeability $UZ \stackrel{d}{=} Z$. Since $\mathcal{G}$ is a group, the orbit $\{gUZ : g \in \mathcal{G}\}$ equals $\{gZ\}$, so $p(UZ)$ is computed from the same multiset of values $\{T(gZ)\}$, and $T(UZ)$ is a uniformly chosen member of it. A uniformly chosen member lies among the top j (counting ties generously) with probability at most $j/|\mathcal{G}|$. Hence $P(p(Z) \leq \alpha) = P(p(UZ) \leq \alpha) \leq \alpha$. Taking expectations over the conditioning gives the unconditional statement. ■

For two samples under $H_0 : F = G$ the pooled observations are i.i.d., hence exchangeable, and the relevant re-assignments are the $\binom{N}{m}$ splits (each realized by $m!\,n!$ permutations). The permutation test is **conditional inference**: the observed values are held fixed and only their labels are random.

3.4 Random (Monte Carlo) permutation tests

When $\binom{N}{m}$ is too large, draw M random relabelings, compute $T_1, \ldots, T_M$, set $T_0 = T_{\text{obs}}$, and use

$$\hat{p} = \frac{1 + \sum_{i=1}^{M} I(T_i \geq T_{\text{obs}})}{M+1}. \tag{3.2}$$

Proposition 3.4

Under H_0, $P(\hat{p} \leq \alpha) \leq \alpha$ for every M. Moreover $\hat{p}$ estimates the exact permutation p-value with standard error about $\sqrt{p(1-p)/M}$.

Proof. Given the multiset of values, $T_0, T_1, \ldots, T_M$ are exchangeable under H_0: the observed labeling is itself a uniform draw, like the M random ones. The rank of T_0 among the $M+1$ values

is therefore uniform (with ties broken against rejection), and $\hat{p} \le j/(M+1)$ iff T_0 is among the j largest. For the second claim, $\hat{p} \approx$ the mean of M Bernoulli(p) indicators. ■

This is why the "+1" is used: it makes the Monte Carlo test exactly valid, not just approximately.

Generating a random relabeling. Draw a uniform random permutation of $1, \dots, N$, and give group 1 either to the units in its first m positions or to the units that receive the first m labels. Either way, each m-subset arises from exactly $m!\,n!$ of the $N!$ permutations, so every split has probability $1/\binom{N}{m}$.

Two-sample alternatives and p-values. With $H_0 : F = G$, the one-sided alternatives are stochastic orderings, $G(x) \le F(x)$ for all x (Y tends to be larger) or the reverse. Under a shift model they become $\Delta > 0$ or $\Delta < 0$. For $D = \bar{Y} - \bar{X}$ with values $d_0 = D_{\text{obs}}, d_1, \dots, d_M$:

$$p_{\text{upper}} = \frac{\#\{d_i \ge d_0\}}{M+1}, \qquad p_{\text{lower}} = \frac{\#\{d_i \le d_0\}}{M+1}, \qquad p_{\text{two}} = \frac{\#\{|d_i| \ge |d_0|\}}{M+1}.$$

With $M = 1999$, the standard error at $p = 0.05$ is about 0.005.

3.5 Other permutation tests

One sample: sign flipping. If $X_i - \theta_0$ are independent and symmetric about 0 under H_0, then $(\pm(X_1-\theta_0), \dots, \pm(X_n-\theta_0))$ have the same distribution for all 2^n sign choices. The permutation test over sign flips with statistic $\sum I(X_i > \theta_0)$ is the sign test. With $\sum V_i R_i$ it is the signed-rank test. With the one-sample t-statistic it is a symmetric-F version of the t-test (so the tests of Chapter 2 are permutation tests).

2×2 tables: Fisher's exact test. With two binary variables (e.g. exposure and disease) and H_0: independence (or equal proportions), hold one variable fixed and permute the other. With margins fixed, the count D in the (exposed, diseased) cell follows the hypergeometric distribution:

$$P(D = d) = \binom{m}{d}\binom{N-m}{k-d}\Big/\binom{N}{k},$$

where m is the number diseased and k the number exposed among N.

Example 3.2

$N = 20$ subjects, $m = 10$ diseased, $k = 6$ exposed, and 5 of the exposed are diseased. Then $P(D \ge 5) = [\binom{10}{5}\binom{10}{1} + \binom{10}{6}\binom{10}{0}]/\binom{20}{6} = (2520 + 210)/38760 = 0.070$.

Summary. Permutation tests (i) need only exchangeability under H_0, (ii) allow any statistic, and (iii) give exact p-values by computation. Their limits: they test only hypotheses that make the data exchangeable (e.g. not "equal means, unequal variances"), and interval estimation requires inverting many tests.

3.6 Selected exercises

Motivation. The exercises quantify the Monte Carlo error of a random permutation test, identify two classical 2×2 statistics with Fisher's test, and derive the moments of the permutation distribution of a mean difference.

Exercise 3.6 (Monte Carlo error of $\hat{p}$)

A random permutation test with $M = 999$ relabelings gave $\hat{p} = 0.01$. Give an approximate 99% confidence interval for the exact permutation p-value p. How much shorter would it be with $M = 1999$?

Solution.

Step 1. Given the data, each random relabeling is uniform over the K splits, so $P(T_i \geq T_{\text{obs}}) = p$ by (3.1). The relabelings are independent, so $\sum_{i=1}^{M} I(T_i \geq T_{\text{obs}}) \sim B(M, p)$ and $\hat{p}$ in (3.2) differs from the binomial proportion only by $O(1/M)$. Hence $\hat{p} \dot{\sim} N(p, p(1-p)/M)$.

Step 2. With $z_{0.005} = 2.576$ and p in the standard error replaced by $\hat{p}$,

$$\hat{p} \pm 2.576\sqrt{\frac{\hat{p}(1-\hat{p})}{M}} = 0.01 \pm 2.576\sqrt{\frac{0.0099}{999}} = 0.01 \pm 0.0081 = (0.0019,\ 0.0181).$$

Step 3. With $M = 1999$ the half-width is $2.576\sqrt{0.0099/1999} = 0.0057$, giving $(0.0043,\ 0.0157)$. The width scales as $M^{-1/2}$, so doubling M multiplies it by $\sqrt{999/1999} = 0.71$, about 29% shorter. Halving the width needs four times as many relabelings. Near a decision boundary such as $p = 0.01$ this matters: with $M = 999$ the interval contains values on both sides of 0.01 by a wide margin. ■

Exercise 3.7–3.8 (χ^2 and odds ratio as permutation statistics)

In the 2×2 setting of Section 3.5 (margins fixed: N subjects, m diseased, k exposed, D = number exposed and diseased), show:
(a) Pearson's χ^2 statistic is an increasing function of $|D - km/N|$, so the permutation test based on χ^2 is the two-sided Fisher test that orders tables by $|D - ED|$;
(b) the sample odds ratio is an increasing function of D, so the permutation test based on it is the one-sided Fisher test.
Apply both to Example 3.2.

Solution. The cells are $a = D$ (exposed, diseased), $b = k - D$, $c = m - D$, $d = N - m - k + D$, with row totals $k, N - k$ and column totals $m, N - m$.

(a)

Step 1. The expected counts under independence are $E_a = km/N$, $E_b = k(N-m)/N$, $E_c = (N-k)m/N$, $E_d = (N-k)(N-m)/N$. Direct subtraction gives

$$a - E_a = d - E_d = D - \frac{km}{N}, \qquad b - E_b = c - E_c = -\Big(D - \frac{km}{N}\Big).$$

Step 2. Hence $\chi^2 = \sum(O-E)^2/E = (D - km/N)^2 \sum 1/E$, and

$$\sum \frac{1}{E} = \frac{N}{k\,m\,(N-k)(N-m)}\big[(N-k)(N-m)+(N-k)m+k(N-m)+km\big] = \frac{N^3}{k\,m\,(N-k)(N-m)}.$$

So $\chi^2 = \frac{N^3}{km(N-k)(N-m)}(D - km/N)^2$. The margins are the same for every permutation, so the constant does not change, and χ^2 is increasing in $|D - km/N|$. Under the null, D is hypergeometric with mean km/N. So the permutation p-value is $P(|D - km/N| \geq |d_{\text{obs}} - km/N|)$.

(b) On the range of D where $b, c > 0$, $\text{OR} = \frac{ad}{bc} = \frac{D}{m-D} \cdot \frac{N-m-k+D}{k-D}$. Each factor is a nonnegative increasing function divided by a positive decreasing one, so OR is increasing in D, strictly where it is positive. At the ends of the range, OR is 0 (if a or d is 0) or $+\infty$ (if b or c is 0), its smallest and largest values. Hence "$\text{OR} \geq \text{OR}_{\text{obs}}$" is the event $D \geq d_{\text{obs}}$, which is the one-sided Fisher test.

In Example 3.2, $km/N = 3$ and $d = 5$. Here $m = N - m = 10$, so $P(D = d) = \binom{10}{d}\binom{10}{6-d}/\binom{20}{6}$ is symmetric about 3. The odds-ratio test gives $p = P(D \geq 5) = 0.070$. The χ^2 test gives $p = P(D \geq 5) + P(D \leq 1) = 0.141$. ■

Exercise 3.9 (Moments of the permutation distribution of $\bar{Y} - \bar{X}$)

Let $z_1, \ldots, z_N$ be the pooled observations, $S^2 = \frac{1}{N-1}\sum_k (z_k - \bar{z})^2$, and let $D = \bar{Y} - \bar{X}$ be computed for a uniformly random split into groups of sizes m (X) and n (Y).
(a) Show that $ED = 0$ and $\operatorname{Var} D = NS^2/(mn)$.
(b) Let $Z = D/\sqrt{NS^2/(mn)}$ and let T be the pooled two-sample t-statistic. Show that $Z^2 = (N-1)T^2/(N - 2 + T^2)$. Hence Z is an increasing function of T, and $Z/T \to 1$ as $N \to \infty$ with T fixed.

Solution [Supplement].

(a) $\bar{Y}$ is the mean of a simple random sample of size n drawn without replacement from $\{z_k\}$, whose population variance is $\sigma^2 = (N-1)S^2/N$. By Theorem 4.2,

$$E\bar{Y} = \bar{z}, \qquad \operatorname{Var}\bar{Y} = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1} = \frac{mS^2}{nN}.$$

Since $m\bar{X} + n\bar{Y} = N\bar{z}$, we have $D = \bar{Y} - (N\bar{z} - n\bar{Y})/m = \frac{N}{m}(\bar{Y} - \bar{z})$. Hence $ED = 0$ and $\operatorname{Var} D = \frac{N^2}{m^2} \cdot \frac{mS^2}{nN} = \frac{NS^2}{mn}$.

(b) From the proof of Proposition 3.2, $(N-1)S^2 = SST = (N-2)s_p^2 + A$ with $A = \frac{mn}{N}D^2$, and $T^2 = A/s_p^2$. Therefore

$$Z^2 = \frac{A}{S^2} = \frac{(N-1)A}{(N-2)s_p^2 + A} = \frac{(N-1)T^2}{N - 2 + T^2}.$$

Z and T have the sign of D, so $Z = g(T)$ with $g(t) = t\sqrt{(N-1)/(N-2+t^2)}$. Its derivative $g'(t) = \sqrt{N-1}\,(N-2)(N-2+t^2)^{-3/2}$ is positive. For fixed t, $g(t)/t \to 1$ as $N \to \infty$.

So the normal approximation to the permutation distribution of D, $Z \dot{\sim} N(0,1)$, and the t-test nearly coincide in large samples. This explains why permutation and normal-theory p-values agree there. ■